
2.

2. ($*$) **Set operations and events, de Morgan's rules** [4 Points]

A (six-sided) die is rolled $N \in \mathbb{N}$ times. Let A_k be the event that the k-th role is a 3 and B_k is the event that the k-th roll is a 6 ($1 \leq k \leq N$).

(a) Express with the help of the events A_k, B_k, as well as elementary set operations the following events:

i. A: 6 is never obtained.

ii. B: a 3 is obtained at least once.

iii. C: a 6 is obtained at least once and a 3 is obtained at least twice.

(b) Describe the following events in words:

$$\left(\bigcup_{i=1}^{N} A_i^c\right)^c, \qquad \bigcup_{i=1}^{N-2}(A_i \cap A_{i+1} \cap B_{i+2}), \qquad \bigcap_{i=1}^{N-1}(A_i \cup B_{i+1}).$$

Now let $(A_i)_{i \in I}$ be a family of subsets of a non-empty set Ω. Prove *de Morgan's rules*:

(c) $\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c$.

(d) $\left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c$.

(a) i. $A = \left(\bigcup_{i=1}^{N} B_i \right)^c$ ii. $B = \bigcup_{i=1}^{N} A_i$

iii. $C = \left(\bigcup_{i=1}^{N} B_i \right) \cap \left(\bigcup_{1 \le i < j \le N} (A_i \cap A_j) \right)$

for (a) and (b), assume N is large enough such that there are no empty union/intersection.

(b)

i. 3 is always obtained

ii. Obtaining the consecutive subsequence 3, 3, 6 at least once.

iii. All 2-element consecutive subsequences have 3 as its 1st element, 6 as its 2nd element, or both.

(c) Assume I is nonempty for (c) and (d).

$$\underbrace{\left(\bigcup_{i\in I} A_i \right)^c}_{L} = \underbrace{\bigcap_{i\in I} A_i^c}_{R}$$

Show $L \subseteq R$

$x \in \left(\bigcup_{i\in I} A_i \right)^c$

$\Rightarrow x \notin \bigcup_{i\in I} A_i$

$\Rightarrow \forall_{i\in I}\ x \notin A_i$

$\Rightarrow \forall_{i\in I}\ x \in A_i^c$

$\Rightarrow x \in \bigcap_{i\in I} A_i^c$

Show $R \subseteq L$

$x \in \bigcap_{i\in I} A_i^c$

$\Rightarrow \forall_{i\in I}\ x \in A_i^c$

$\Rightarrow \forall_{i\in I}\ x \notin A_i$

$\Rightarrow x \notin \bigcup_{i\in I} A_i$

$\Rightarrow x \in \left(\bigcup_{i\in I} A_i \right)^c$

$L \subseteq R$ and $R \subseteq L \Rightarrow L = R$

(d)

(d) $\left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c$

equivalent to

$\left(\bigcup_{i\in I} A_i^c\right)^c = \bigcap_{i\in I} A_i$

Then identify $B_i = A_i^c$

$\left(\bigcup_{i\in I} B_i\right)^c = \bigcap_{i\in I} B_i^c$

This is of the same form as (c), so the sure proof applies.

3.

3. (∗) **Rolling dice** [4 Points]

A fair die is rolled twice. Calculate the probability for the following events.

(a) The number 6 occurs exactly once.

(b) Both numbers are odd.

(c) The sum of the numbers is 4.

(d) The sum of the numbers is an integer multiple of 3.

Write down the probability space $(\Omega, \mathscr{F}, \mathbf{P})$ and the corresponding events explicitly.

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for all subquestions,

$$\Omega = \{1, \dots 6\} \times \{1, \dots, 6\}$$

$$\mathcal{F} = P(\Omega) \quad \text{(powerset of } \Omega\text{)}$$

$$P(A) = \frac{|A|}{36} \quad A \in \mathcal{F}$$

(a) event $= \{(a,b) \in \Omega : (a=6 \text{ or } b=6) \text{ and } ((a,b) \neq (6,6))\}$

event card $= 6 + 6 - 1 - 1 = 10$

prob $= \frac{10}{36} = \frac{5}{18}$

(b) event $= \{1,3,5\} \times \{1,3,5\}$

event card $= 3 \cdot 3 = 9$

prob $= \frac{9}{36} = \frac{1}{4}$

(c) event $= \{(a,b) \in \Omega : a+b=4\}$

event card $= 3$

prob $= \frac{3}{36} = \frac{1}{12}$

(d) event $= \{(a,b) \in \Omega : a+b \in \{3,6,9,12\}\}$

event card $= 2+5+4+1 = 12$

prob $= \frac{12}{36} = \frac{1}{3}$